Evaluation of a Community-Based Participatory Model of Family Education

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Introduction

Family Academy and the NAZ

Family Academy (FA) is a promising parent education model for low-income, primarily African-American, parents of children 0-3 years old who live in a geographic zone in Minneapolis characterized by low school readiness rates, a wide educational achievement gap, high poverty, and high rates of family and community violence. It is a unique model, developed through a grassroots community-engaged process with participants, practitioners and researchers.

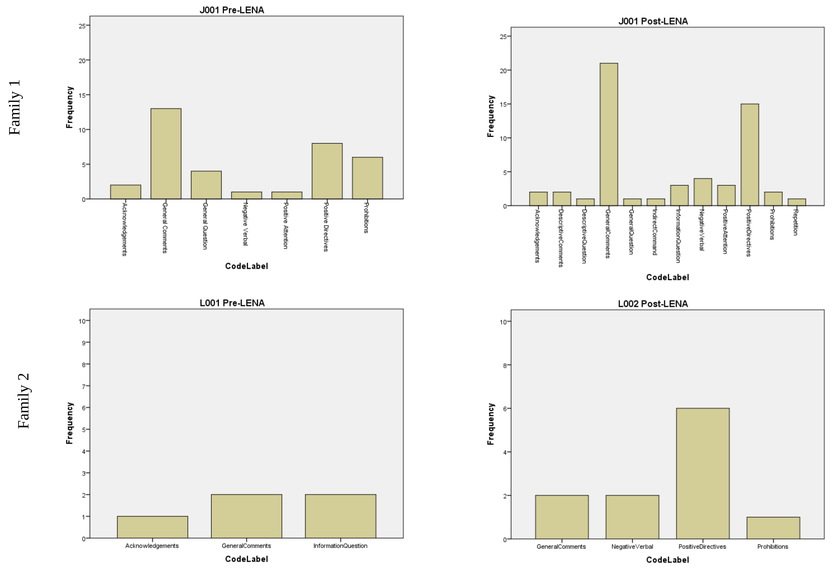
FA is the early childhood parent education component of the Northside Achievement Zone (NAZ) of Minneapolis, a zone-based approach to building a continuum of educational and social support for children, birth to age 18, and their families, ensuring all children in NAZ are college-ready for college. In December 2011, NAZ was awarded a 28 million dollar Promise Neighborhoods grant from the federal government to support expansion and evaluation of its model. Family Academy plays a large role in this expansion, as it is the most prominent evidence-based support offered to families by the NAZ, and as such, NAZ is highly invested in its success.

CBPR Methodology

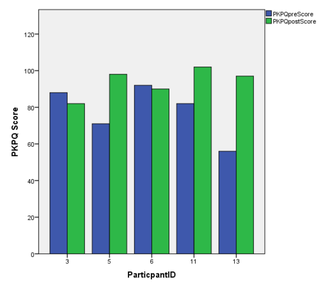
The evaluation of Family Academy examines change in parent knowledge and practices that closely aligns with curricular components. This measurement model diverges from the traditional focus on distal measures of the target construct, to specifically evaluate the degree to which parents receiving the intervention actually demonstrate increased use of the behavior within their daily lives (Slavin & Madden, 2011). Further, as this project is strongly rooted in a community-based participatory approach (Minkler & Wallerstein, 2011), NAZ program staff were an integral component of measure and measurement model development, which further strengthens the construct validity of the developed measures. Specifically, during a process of iterative development, NAZ program staff provided direct input on the design and wording of items and data collection methodology to increase the likelihood that the measures were culturally relevant and solicited information in a meaningful and valid way from participants.

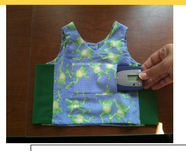
Research Question

Do measures developed by the research team; the Parent Language Behavior Code (PLBC) and the Parent Knowledge and Practices Questionnaire (PKPQ) detect change in parent knowledge and skills when administered before and after the Family Academy intervention is delivered?



PKPQ Pre- and Post Data





Measures: PKPQ and LENA



LENA

The Language ENvironment Analysis (LENA) is a cutting-edge technology system used to gather and analyze data in a child's home language environment via audio recordings. It consists of a digital language processor (DLP) and speech recognition software. Children wear the DLP in the pocket of custom-made clothing and the system records all audio data occurring in the home for up to 16 hours. LENA recordings were conducted twice during the Family Academy session.: once within the two week period prior to the start of the Family Academy program and once within a month of the end of Family Academy.

Of the 16 hour audio file, three 10 minute segments of audio data (with parameters for sampling standardized across participants) were selected, transcribed, and coded using the Parent Language Behavior Code (PLBC), developed by the research team. The PLBC has been developed to capture the frequency with which specific language behaviors directly taught within the Family Academy curriculum are occurring in the home environment.

PKPQ

The Parent Knowledge and Practices Questionnaire (PKPQ) consists of approximately 40 items that are either true/false or short answer which reflect parent's level of knowledge on positive parenting practices specifically taught by the Family Academy Curriculum. The PKPQ also includes rating scale items that ask parents to rate their typical use of specific parenting practices.

References

Minkler, M., & Wallerstein, N. (Eds.). (2011). *Community-based participatory research for health: From process to outcomes*. Jossey-Bass.

Slavin, R. & Madden, N. (2011): Measures Inherent to Treatments in Program Effectiveness Reviews, Journal of Research on Educational Effectiveness, 4:4, 370-380

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